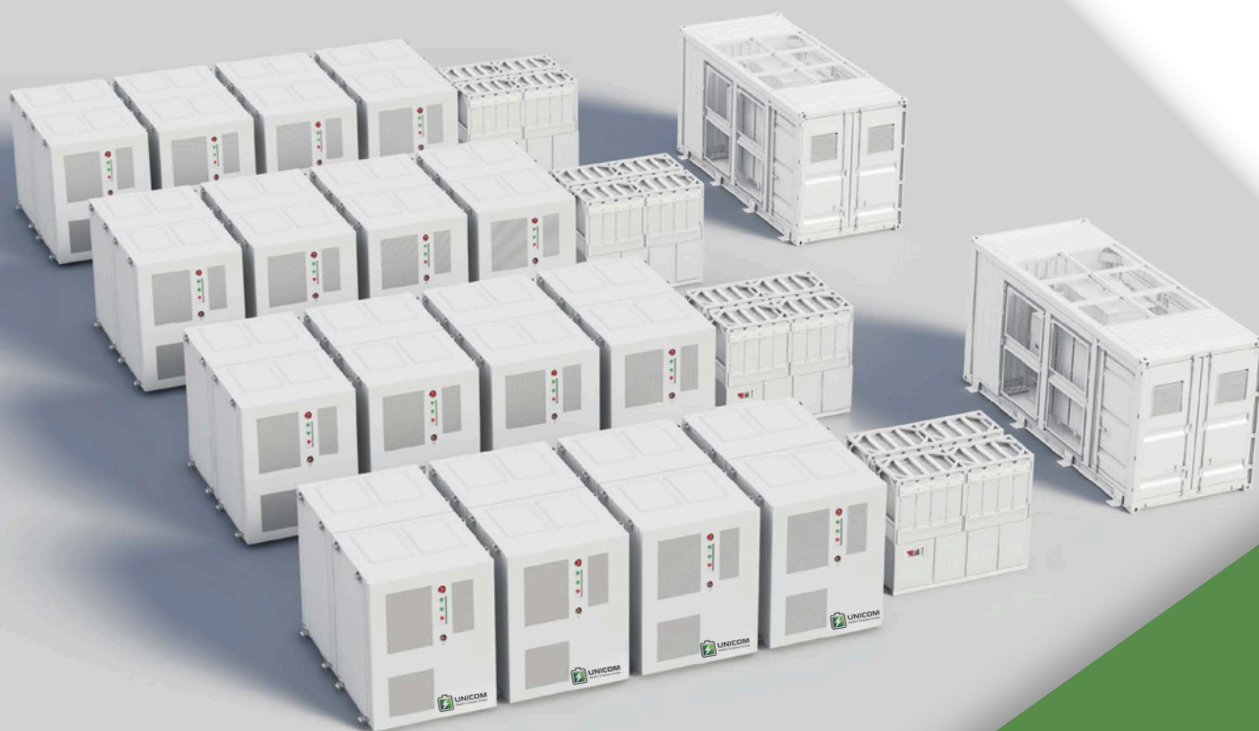




UNICOM
ENERGY STORAGE SYSTEM



FlexCombo

Flexible, Scalable & Reliable Energy Block

FlexCombo



FlexCombo is a reliable, intelligent and fully-integrated battery energy storage system developed by Unicom Energy Storage System for applications in wide-range projects conditions and locations.

Modular design & parallel-connected method of FlexCombo indicate more flexibility and effortless-experience by shortening the delivery period and simplifying the deployment process.



All-in-One & Plug-and-play
Ease of transportation, installation, operation and maintenance;



Flexible Scalability
Simply deploying & connecting standardized blocks to increase energy capacity;



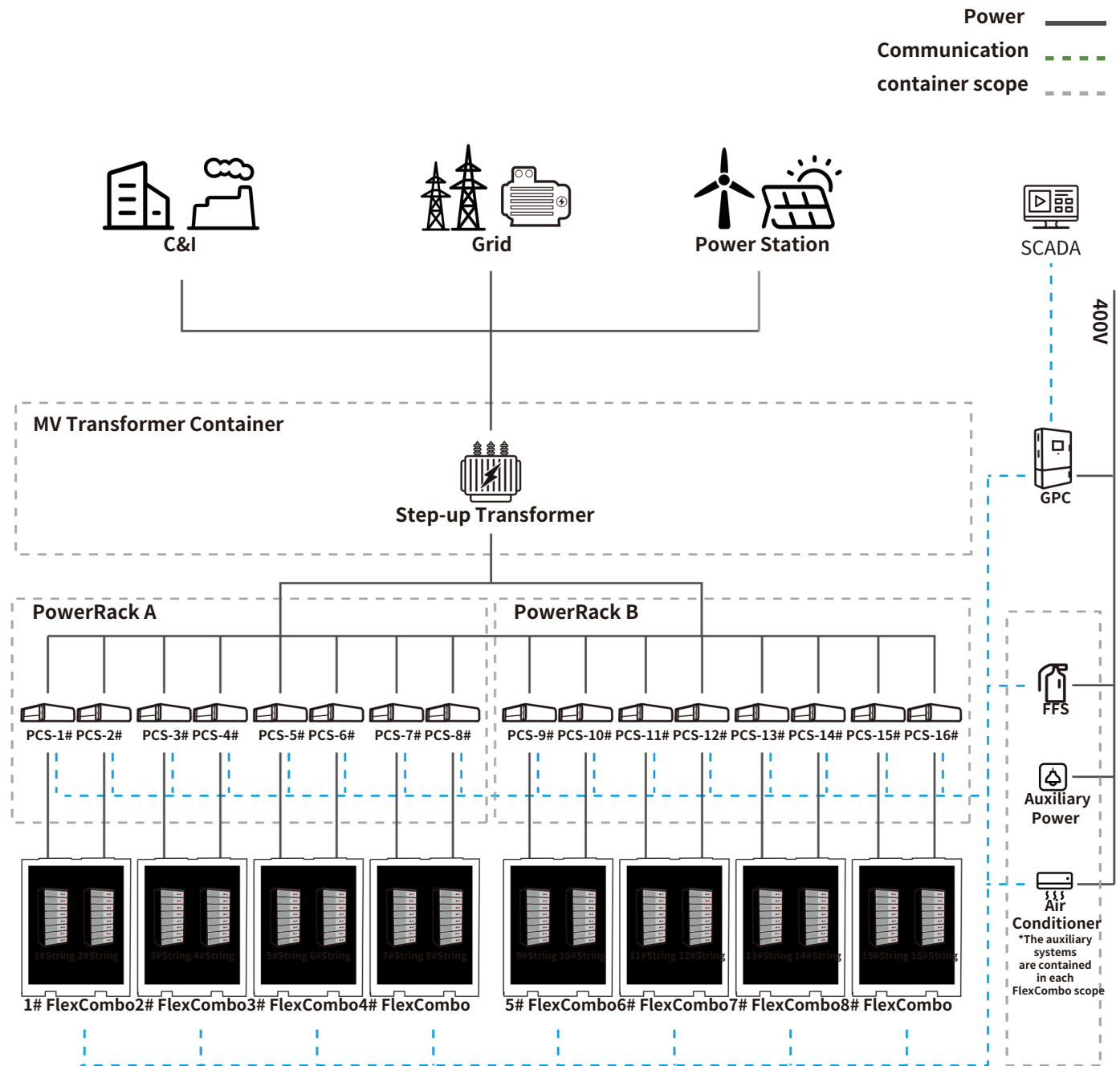
Intelligent Management
Integrated with leading balancing technology & operation platform, huge plus for safety and long-life-span.



FlexCombo is designed with an easy augmentation feature to adapt evolving project requirements. Its DC/AC one-on-one topology, combined with a modular design, simplifies AC augmentation by enabling the addition of PCS units and new enclosures as needed.

This flexible approach supports incremental expansion of the system. By integrating two standardized FlexCombo units and two additional PCS to at each stage, the system's capacity can be scaled up from 6,687kWh, ensuring at least 10 years of stable projects operation.

System Topology



Product Configuration

Product Model	Battery String Type	FlexCombo Qty	Nominal Capacity	DC Voltage Range	PCS Grid-connected Voltage
FlexCombo - D6500	LS417	8	6,687kWh	1,164.8V~1,476.8V	690V

System Technical Specifications

Item	FlexCombo-D6500	
DC Data		
Battery chemistry	Lithium Iron Phosphate (LFP)	
Number of D1 (DC block) DC rated energy capacity	8	
RCaettle sdp veocltag	6,687kWh	
Voltage range	1,331.2V	
Auxiliary load voltage	1,164.8V~1,476.8V	
Auxiliary peak load	400V, 50/60Hz	
BMS communication	54kW	
interface	RS485, Ethernet	
BMS communication protocol	Modbus RTU, Modbus TCP	
AC Data		
Rated AC Power	3,200kW	3,520kW
Maximum AC power	690V	586.5~759V
Rated voltage	(Optional)	2,677.6A
GCerildl svpoeltcage range	<3% -1~+1 3W+PE	
AC rate of current		
Output THDi		
AC PF		
AC output		
General Data		
Degree of protection	IP54	
COeplle rsaptiencg	-30~50°C	
temperature range	0~95% (non-condensing)	
Relative humidity	<2,000m/6,561ft	
Max working altitude	Liquid Cooling	
Cooling concept of DC hatch	50% Ethylene glycol solution	
CCoeolll sapnet ctype	Aerosol/Water fire fighting	
Fire fighting system	RS485, Ethernet	
Communication interfaces	UL9540, CE	
Certificates		

Key Components



Flexcombo

Item	D1
Battery chemistry	Lithium Iron Phosphate (LFP)
Cell life cycle	8,000 cycles with 70% retention @ 0.5C 25°C
Cell spec	3.2V/314Ah
Each string configuration	1P416S
CNeulml sbpeer cof strings	2
DC rated energy capacity	836kWh
Rated voltage	1,331.2V
Voltage range	1,164.8V~1,476.8V
Auxiliary load voltage	400V, 50/60Hz
Auxiliary peak load	6.7kW
BMS communication interface	RS485, Ethernet
BMS communication protocol	Modbus RTU, Modbus TCP
Dimension (W*D*H)	2,000mm*1,785mm*2,545mm
Weight of the whole system	7.8t

□ Key Components



Power Conversion System

Iteem	EH-0200-HA-M
Rated AC power	200kW 224.5A
Max.DC current	1000~1500Vd.c
DC voltage range	1500Vd.c
Max.DC voltage	-1~+1
Power factor range	IP66
Ingress protection	167.4A
Rated AC current	184.1A
Max.AC current	690Va.c.(3W+PE)
Rated AC voltage	-15%~+10%(Configurable)
Output operating voltage range	50/60Hz
Rated AC frequency	-40~+60°C
Oprating temperature range	I
Protective class	4000m(Derating above 3000 m)
Operating altitude	
Cooling method	Smart air cooling
Weight	100kg



PowerRack-EU

Iteem	
DC Parameters	
Max.DC current	224.5A*8
DC voltage range	1,000-1,500V
Number of DC inputs	8
AC parameters (On-Grid)	
Rated output power	200kW*8
Rated grid voltage	690V, 3W+PE
Output operating voltage range	-15%~+10%(Configurable)
Grid frequency	50Hz/60Hz
Rated AC current	167.4A*8
AC PF	-1~+1
Dimensions (W*D*H)	2,550mm*900mm*2,100mm
Weight	≤2t

- Single-stage three-level modularization;
- Multi-branch input to reduce battery series and parallels connection;
- One-to-one Management of Battery string and PCS.

□ Key Components



Certificates: IEC 62271-202, EN 50588-1, IEC 60076, IEC 62271-200, IEC 61439-1

MV Transformer

Item	CBMV-3300-EU	CBMV-6600-EU
Input Rated Input Voltage Phase number AC Power LV Main Switches	690V	
Output Rated Output Voltage	3φ	
Frequency Transformer Tappings	3300kVA@40°C	6600kVA@40°C
Transformer Cooling Type	ACB (1,600A/3P) × 2	ACB (1,600A/3P) × 4
Transformer Oil Type		
Transformer Vector Group	10~35kV	
Transformer Min. Peak Efficiency	50Hz/60Hz	
Index	± 2 x 2.5%	
HVC Type HVC Transformer Protection	ONAN	ONAN
Unit Transformer Type	Mineral Oil (PCB Free)	Mineral Oil (PCB Free)
Auxiliary Transformer type	Dy11	Dy11-y11
Output Voltage of Auxiliary Transformer	Tier 2 In Accordance with EN 50588-1	
General	SF6 Gas Insulated	
Dimensions (W*D*H)	MV Vacuum circuit breaker	
Weight	Oil-immersed Transformer	Oil-immersed Transformer
Protection Degree of MV & LV Room	SGL-100kVA 0.69/0.4kV Dyn11	SGL-100kVA 0.69/0.4kV Dyn11
Operating Temperature Range	400Vac	
Relative Humidity		
Operating Altitude	6,058mm*2,438mm*2,896mm	6,058mm*2,438mm*2,896mm
MV-LV AC Connections	<16t	<25t
LV & MV Room Cooling	IP 54	
Communication protocols	-30°C~40°C	
	0~95% (Non-condensing)	
	1,000m	
	Prewired and Pretested, No Internal Cabling Onsite	
	Forced air cooling	
	Modbus TCP/IP	

□ Next-Gen Control



GridPoint Controller (rGPC)
an advanced one-stop BESS control and status information and management center.

Power interface
AC400V/DC24V

Communication
Modbus TCP

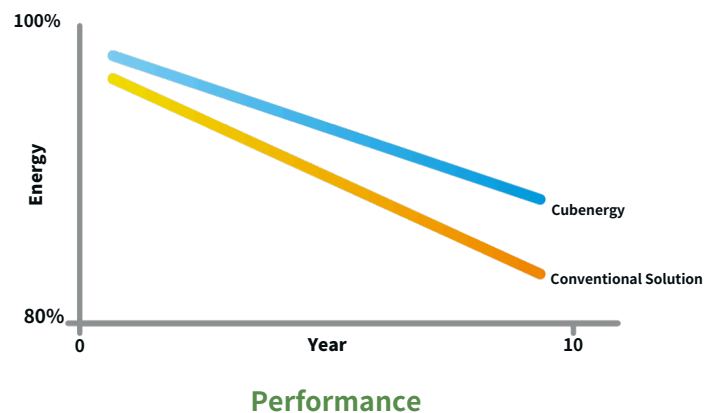
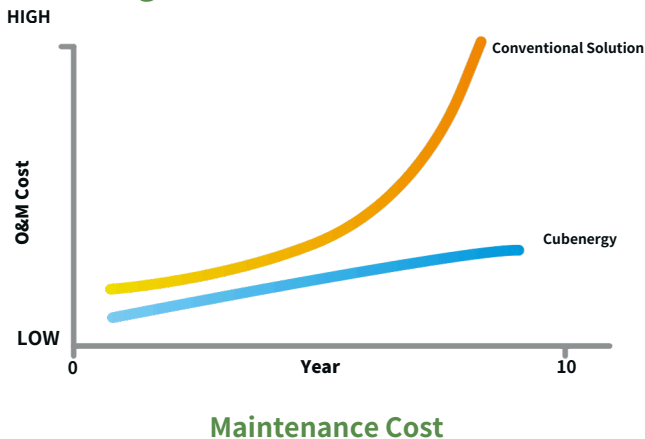
SPAC BMS

SPAC BMS delivers intelligent battery management through real-time cell voltage balancing. This proprietary technology minimizes energy loss caused by cell imbalances, increasing balance efficiency and extending battery cycle life. By ensuring uniform charge/discharge distribution across all cells, SPAC BMS maximizes usable capacity while maintaining stringent safety protocols—critical for large-scale BESS applications requiring long-term reliability.

Functions

1. Monitor and protect safety of battery cell
2. Monitor and protect safety of electric system
3. Calculate and manage charging and discharging status
4. Calibrate and manage available energy
5. Optimize balance and manage cell consistency

Advantages



□ Grid Forming*

Unicom's BESS is equipped with advanced technologies for grid voltage support and frequency regulation. It employs Virtual Synchronous Generator (VSG) control to emulate the external characteristics of synchronous generators. This delivers essential inertia, damping, and dynamic grid support, ensuring robust and stable power system operation.



Superior Grid Compliance



Maximized ROI



Enhanced Grid Resilience

Functions

- **Inertia and Damping Control Inertia Emulation:** The inverter simulates inertia using a virtual inertia time constant, allowing it to release or absorb energy during grid frequency or voltage changes, helping to slow down system fluctuations.
Damping Control: A virtual damping factor is applied to suppress oscillations and improve the dynamic stability of the grid.
- **Autonomous Frequency/Voltage Regulation**
Frequency Control: Dynamically adjusts active power based on frequency deviation (parameter kf: 0-200).
Voltage Control: Dynamically regulates reactive power based on voltage deviation (parameter kv: 0-100).
- **Multi-Scenario Adaptability**
Islanded Operation: Maintains voltage/frequency independently during grid outages for critical load supply.
Weak Grid Support: Stabilizes voltage/frequency in high-renewable penetration scenarios.
Renewables Coordination: Optimizes operation with wind/PV through power sharing.

Advantages



- **Enhanced Grid Stability** Compensates for the lack of inertia in renewable energy sources (PV/wind), mitigates intermittent power fluctuations from PV/wind generation, improves grid absorption capacity and suppresses frequency deviations.



- **Flexible Operation Capability**
Supports both grid-connected and islanded modes, maintaining voltage/frequency in microgrids, and participates in grid frequency/voltage regulation in grid-tied application.



- **Multi-unit Cooperative Control** Enables precise power allocation among parallel inverters via virtual impedance and adaptive algorithms, solving uneven power distribution issues in traditional droop control.



- **Power Quality Optimization**
Reduces current THD (Total Harmonic Distortion) to solve voltage imbalance issues caused by PV inverter grid integration.

*Seamless PQ Mode (Fixed Power Output) and GFM Mode (VSG Mode) Switching for Multi-Scenario Operation.